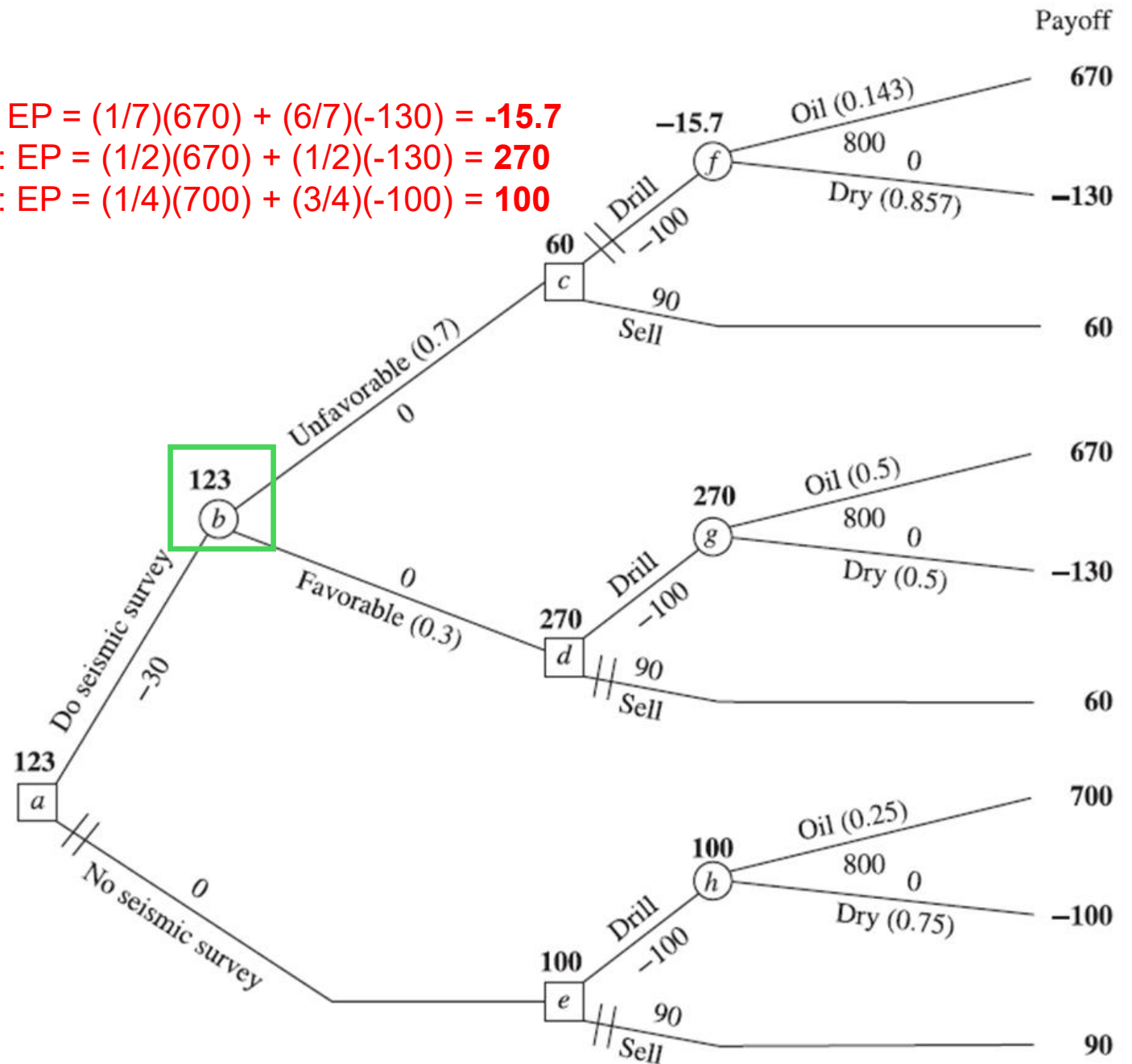


Decision Tree and Utility Theory

Decision Tree

- Node f: $EP = (1/7)(670) + (6/7)(-130) = -15.7$
- Node g: $EP = (1/2)(670) + (1/2)(-130) = 270$
- Node h: $EP = (1/4)(700) + (3/4)(-100) = 100$

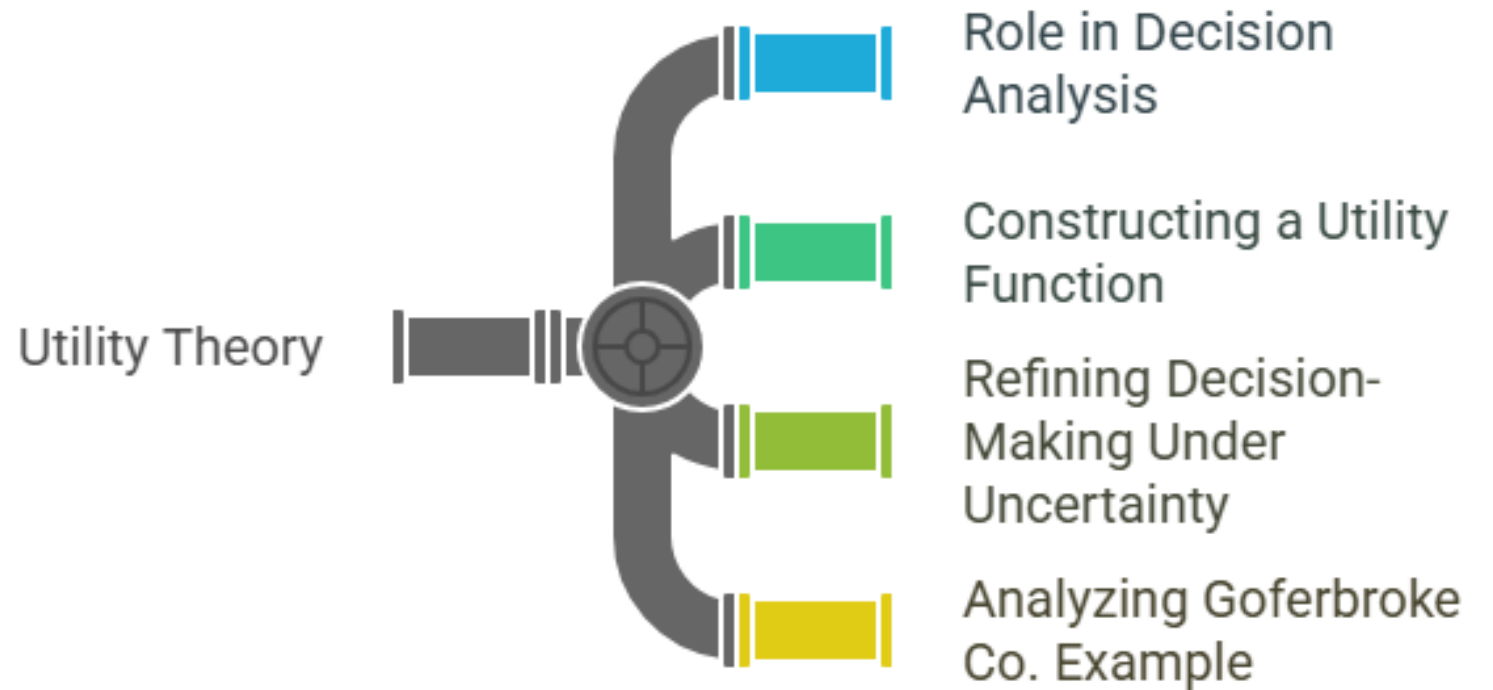


- Start at the **right side** of the decision tree and move **left** one column at a time.
- For each column:
- **Event Nodes:** Calculate expected payoff (EP) as (probability × payoff).
- **Decision Nodes:** Choose the branch with the largest EP; mark rejected branches with "//".

Utility Theory in Decision Analysis

Objectives

Exploring Utility Theory in Decision-Making



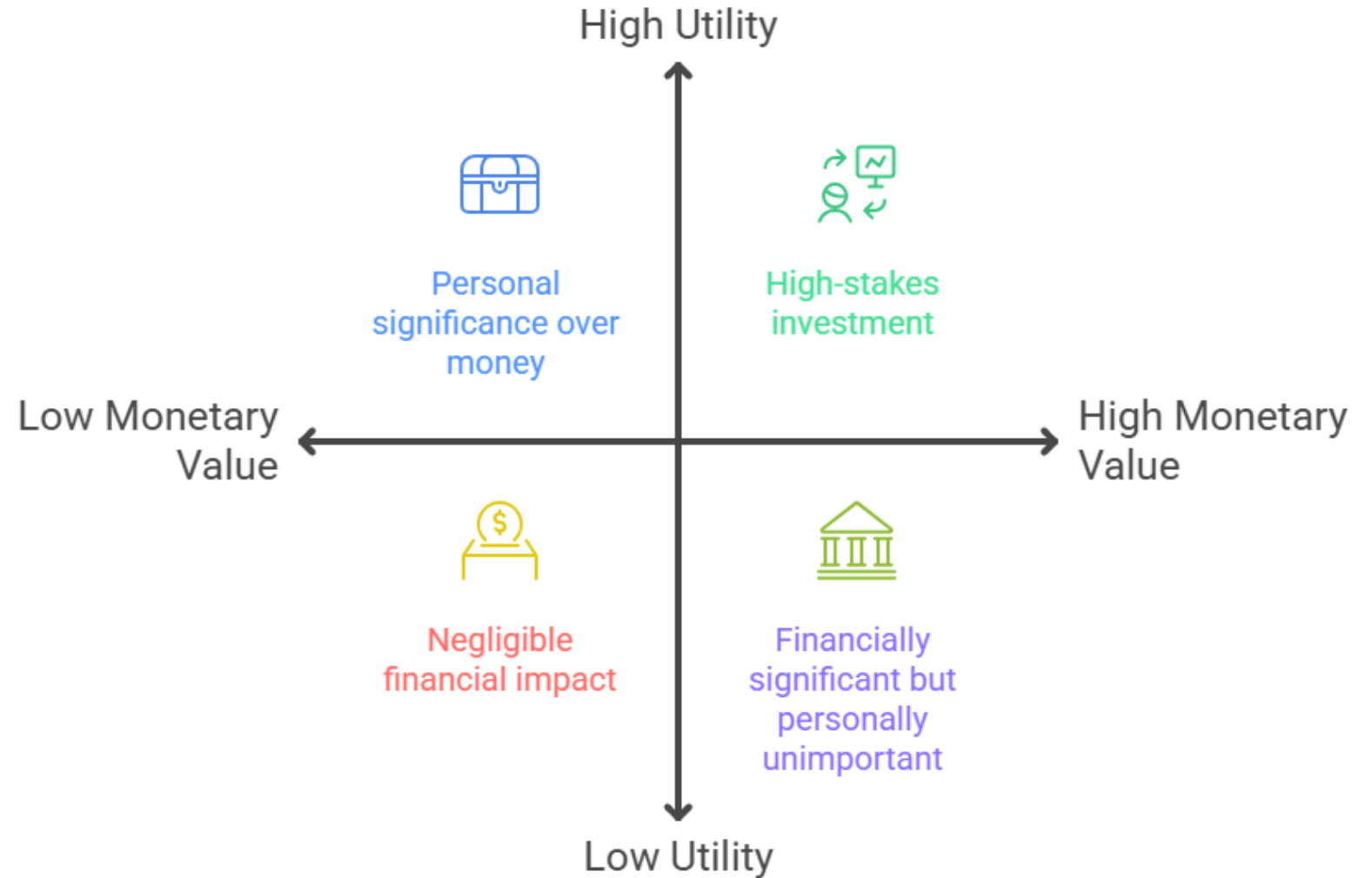
Why Utility Theory?

- Decision analysis often uses expected **monetary payoffs** (Bayes' Decision Rule).
- Problem: Money doesn't always **reflect true value or risk tolerance**.
- Utility Theory: Assigns a numerical value (utility) to outcomes based on the decision maker's preferences and risk attitude.
- Example: \$100 loss might hurt more than a \$100 gain feels good.



Utility framework

Utility vs. Monetary Value in Decision Making

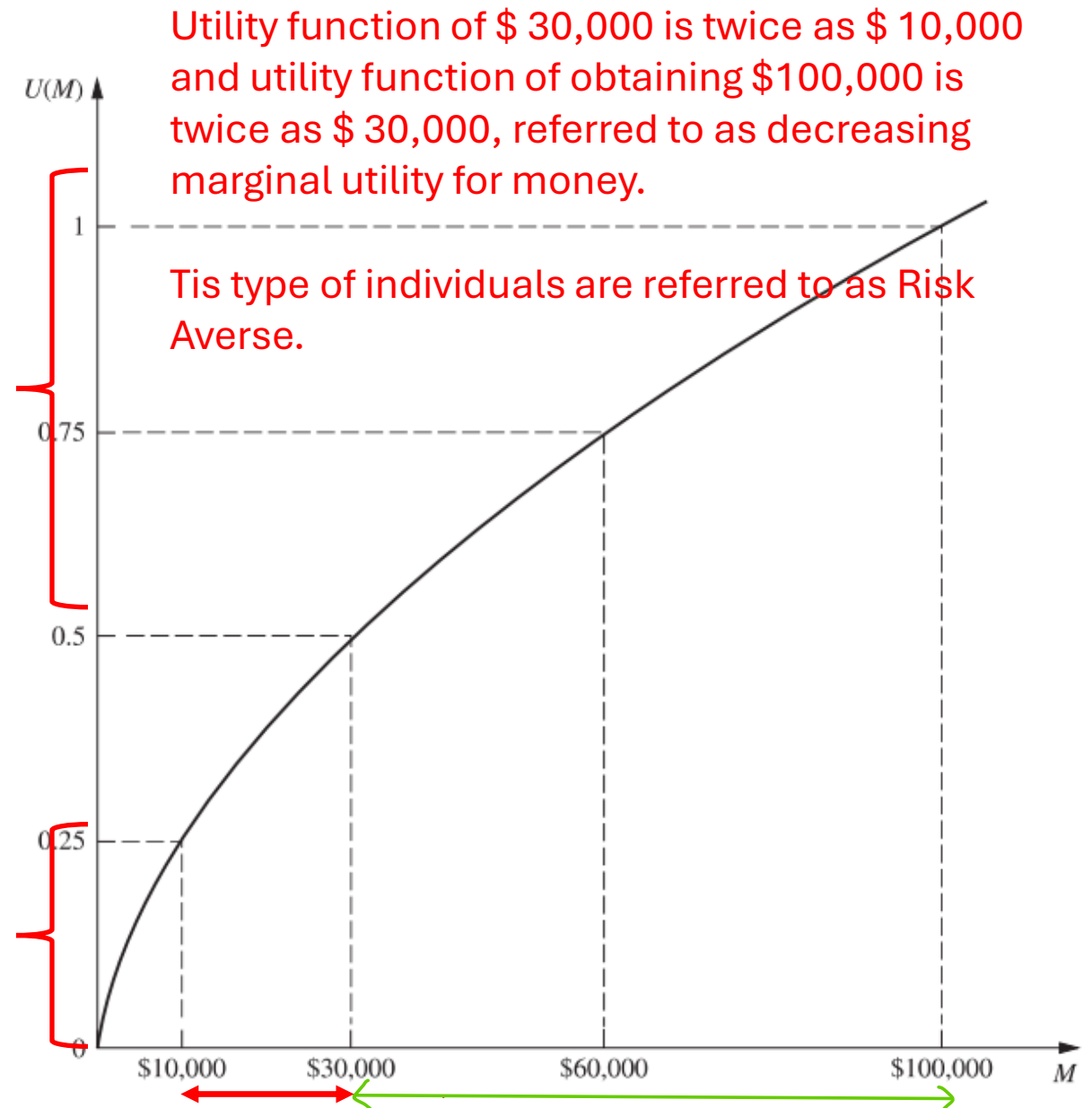


Core Ideas of Utility Theory

- Utility Function ($U(x)$): Maps monetary payoffs (x) to utility values.
- Risk Attitudes:
 - Risk-Averse: Prefers certainty over a gamble with equal expected value.
 - Risk-Neutral: Focuses on expected monetary value.
 - Risk-Seeking: Prefers a gamble over certainty.
- Utilities replace payoffs in decision analysis.

A Typical Utility Function of Money

■ **FIGURE 16.7**
A typical utility function for money, where $U(M)$ is the utility of obtaining an amount of money M .



Constructing a Utility Function “How to Build $U(x)$ ”

- Choose Reference Points: Assign $U(\text{min payoff}) = 0$, $U(\text{max payoff}) = 1$.
- Assess Intermediate Points: Use hypothetical choices between certain payoffs.
- Example: What probability p makes you indifferent between \$500 for sure and a lottery with 50% chance of \$1,000 and 50% chance of \$0?
- If $p = 0.6$, then $U(500) = 0.6 \times U(1000) + 0.4 \times U(0) = 0.6 \times 1 + 0.4 \times 0 = 0.6$.
- Fit a curve to the points.

Revisiting Goferbroke Co.

- Problem: Decide to drill for oil or sell land, with or without a seismic survey.
- Payoffs (in \$000s):
 - Drill: \$700 (oil), -\$100 (dry)
 - Sell: \$90 (oil or dry)
 - Seismic survey costs \$30.
- Bayes' Decision Rule: Drill without survey (expected payoff = \$100,000).
- But the issue is: Company is low on capital; a \$100,000 loss is devastating.

Goferbroke Co.: A High-Stakes Decision

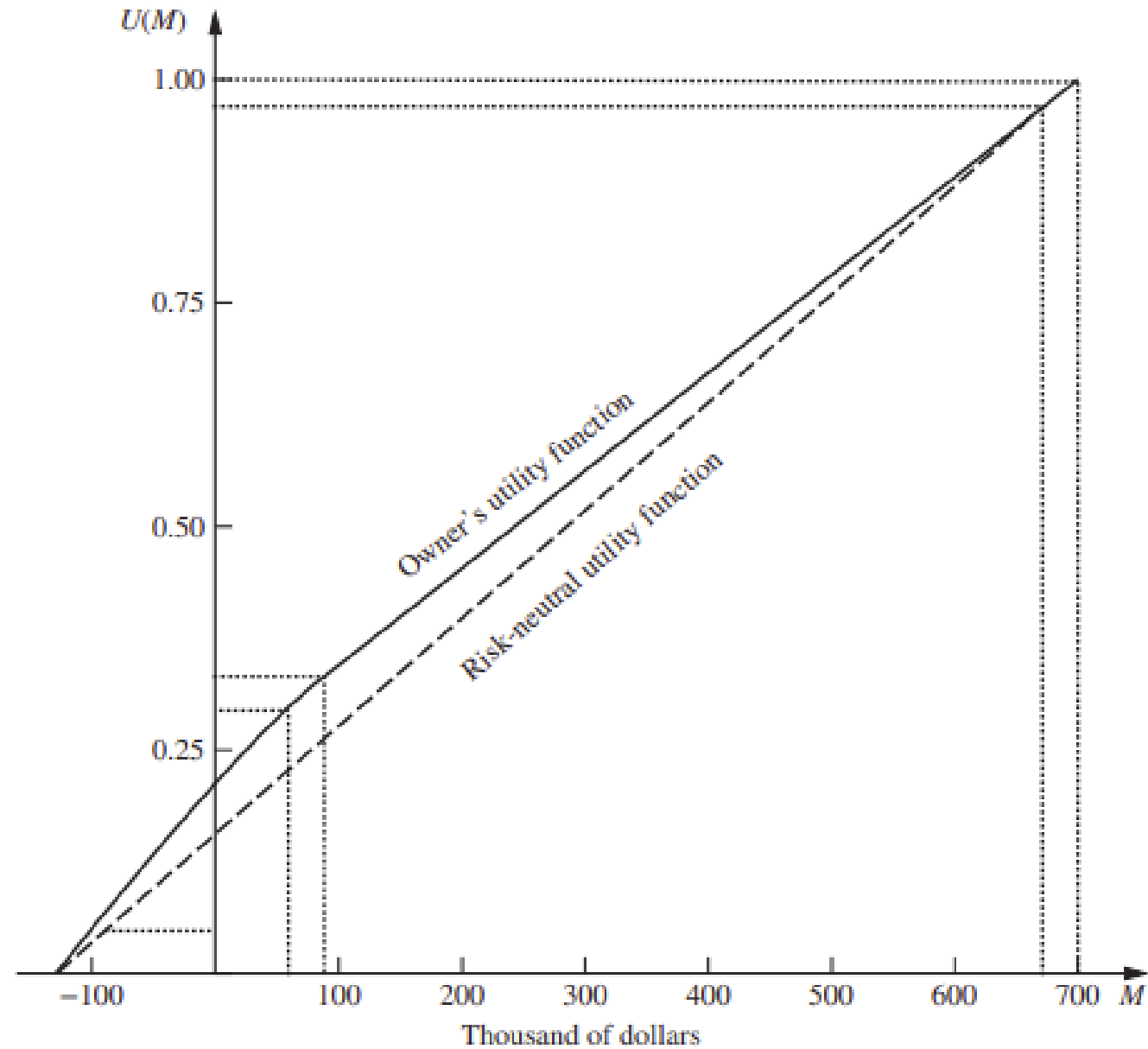
- Operating with limited capital, heavily in debt.
- **Worst-case scenario:**
 - Spend \$30K on seismic survey + lose \$100K drilling (no oil).
 - Outcome: Precarious financial position (not bankrupt yet).
- **Best-case scenario:**
 - Strike oil, earn \$700K → Solid financial footing.
- Owner's stance: Naturally risk-seeking but forced to be moderately risk-averse due to circumstances.

Why Utility Theory?

- Monetary payoffs alone don't reflect the owner's risk perception.
- Utility function: Translates money (M) into subjective value ($U(M)$).
- Owner's profile:
 - Moderately risk-averse (curve bends below risk-neutral line).
 - Contrast: Risk-neutral = straight 45° line (Fig. 16.8 dashed).

Utility Function for money

■ **FIGURE 16.8**
The utility function for
money of the owner
of the Goferbroke Co.



Building Utility Function

- **Step 1.** Set arbitrary numbers
 - $U(-130) = 0$ (worst outcome).
 - $U(700) = 1$ (best outcome).
- **Step 2:** Use equivalent lottery method:
 - Compare risky payoff vs. certain payoff.
- Example questions to owner:
 - $M = 90$: $p = 1/3 \rightarrow U(90) = 0.333$.
 - $M = -100$: $p = 1/20 \rightarrow U(-100) = 0.05$.

Utility is not calculated. It is “revealed” through choices under uncertainty.

Building Utility Function (Conti...)

- Step 3: Draw smooth curve through points:
 - $U(-130) = 0$, $U(-100) = 0.05$, $U(90) = 0.333$, $U(700) = 1$.
- Step 4: Interpolate remaining values:
 - $U(60) = 0.30$, $U(670) = 0.97$.

■ **TABLE 16.7** Utilities for the full
Goferbroke Co. problem

Monetary Payoff	Utility
-130	0
-100	0.05
60	0.30
90	0.333
670	0.97
700	1

Interpolation of values

Example: $U(60)$

Use points:

- $(x_1, U_1) = (-100, 0.05)$
- $(x_2, U_2) = (90, 0.333)$

Formula: $U(x) = U_1 + \frac{x-x_1}{x_2-x_1} (U_2 - U_1)$

Substitute:

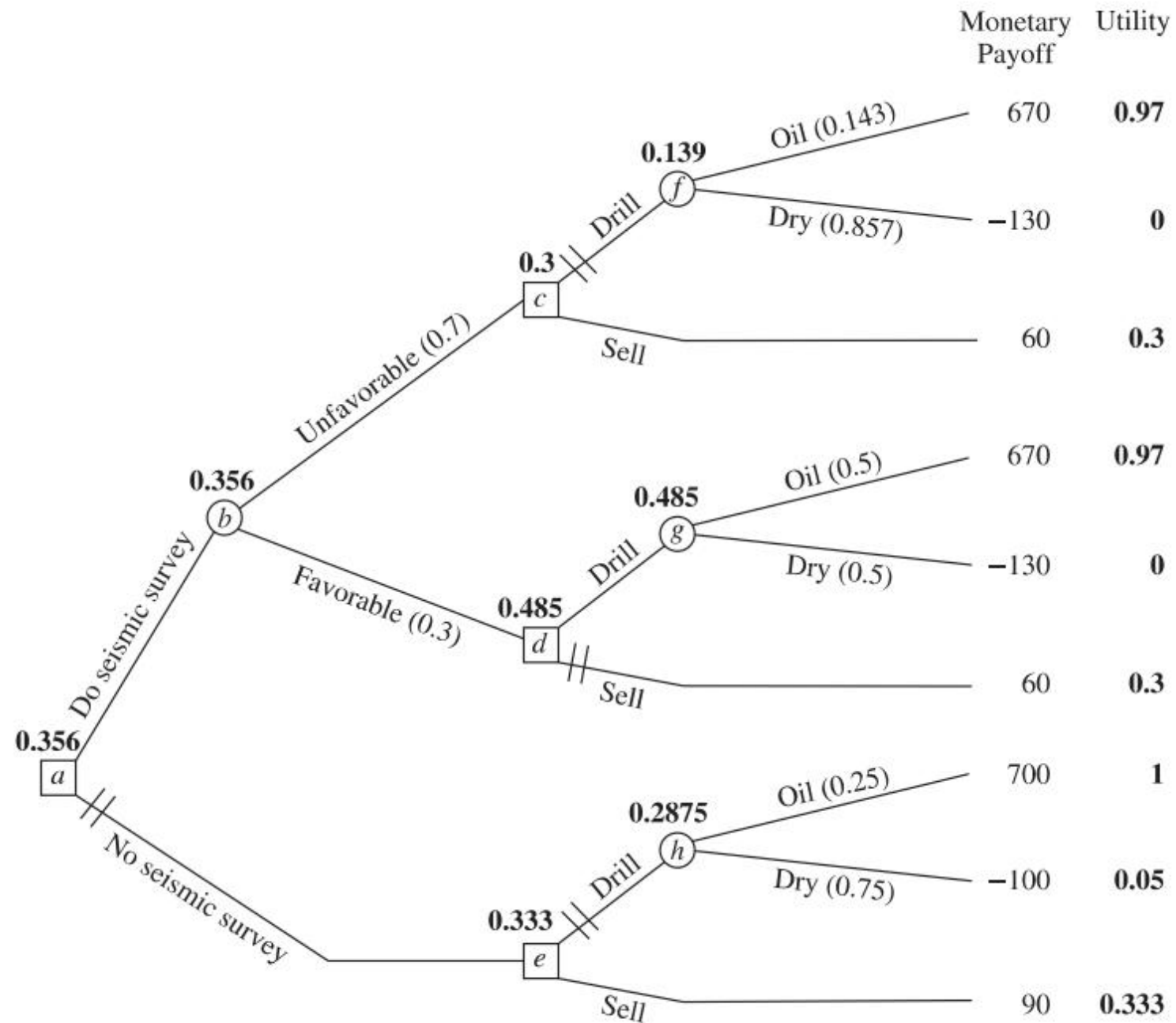
$$\begin{aligned} U(60) &= 0.05 + \frac{60 - (-100)}{90 - (-100)} (0.333 - 0.05) \\ &= 0.05 + \frac{160}{190} \times 0.283 \\ &\approx 0.05 + 0.238 = 0.288 \approx 0.30 \end{aligned}$$

How to Interpret?

- Captures owner's risk attitude under financial strain.
- Outcome:
 - Worst case (-130) has zero utility → Avoid at all costs.
 - Best case (700) has max utility → Highly desirable but tempered by risk aversion.

Decision Tree with Utilities

- Replace monetary payoffs in Fig. 16.5 with utilities from Table 16.7.
- Use same probabilities (prior and posterior) from earlier sections.
- Compute expected utility at each node instead of expected payoff.



■ **FIGURE 16.9**
The final decision tree for the full Goferbroke Co. problem, using the owner's utility function for money to maximize expected utility.

Backward Induction with Utilities

- Terminal Nodes: Assign utilities (e.g., $700 \rightarrow 1$, $-100 \rightarrow 0.2$).
- Event Nodes: Calculate expected utility using probabilities.
 - Example: Drill, no survey: $EU = 0.25 \times 1 + 0.75 \times 0.05 = 0.4$.
- Decision Nodes: Choose the branch with the highest expected utility.
 - Result: Sell without survey ($EU = 0.7$) beats drill ($EU = 0.4$) or survey options.

Optimal Policy with Utilities

- Optimal Decision: Sell the land without a survey ($EU = 0.7$).
- Contrast with Bayes': Drill without survey (expected payoff = \$-100,000).
- Why Different?: Utility reflects risk aversion; avoids the devastating -\$100,000 loss.
- Key Insight: Utilities align decisions with the decision maker's true values.

Sensitivity Analysis (Testing Robustness)

- Vary prior probability of oil (p) from 0.15 to 0.35 (as in Sec. 16.2).
- Recalculate expected utilities:
 - Drill: $EU = p \times 1 + (1-p) \times 0.2 \rightarrow 0.2$ is the utility of dry well
 - Sell: $EU = 0.7 \rightarrow$ Utility of selling (certain payoff)
- Crossover: $p \times 1 + (1-p) \times 0.2 = 0.7 \rightarrow p = 0.625$.
- Conclusion: Sell unless oil is very likely ($p > 0.625$).

Utility Theory in the Real World (Practical Applications)

- Personal Finance: Insurance vs. investment choices.
- Business: Product launches with high risk vs. steady returns.
- Healthcare: Treatment options balancing cost, risk, and quality of life.
- Example: Polio eradication (vignette) used utilities in large decision trees.

Key Messages

- Utility theory refines decision analysis by incorporating preferences and risk.
- Construct $U(x)$ using reference points and indifference probabilities.
- Apply via decision trees with expected utilities.
- Goferbroke Co.: Risk aversion shifts decision from drilling to selling.

Practical Applications of Decision Making

Overview of Decision Analysis in Practice

- Purpose: Provides a structured approach to decision-making under uncertainty
- Key Tools: Payoff tables, decision trees, utility theory, Bayes' decision rule
- Real-World Impact: Optimizes choices in business, government, and beyond
- Example: Goferbroke Co. – Drill or sell land?

Decision Analysis in Business

- Optimize portfolios under economic uncertainty
- Evaluate costs and competition (e.g., government contracts)
- Benefits:
 - Quantifies trade-offs
 - Incorporates probabilities and utilities
 - Sensitivity analysis for robust decisions

Steps for Practical Implementation

- Step 1: Define alternatives and states of nature
- Step 2: Build payoff tables or decision trees
- Step 3: Assign probabilities (prior/posterior)
- Step 4: Evaluate with Bayes' rule or utility functions
- Step 5: Conduct sensitivity analysis or experimentation
- Key Insight: Balances cost, risk, and reward